

- c) $q \rightarrow \neg p$: If you miss the final examination, ^{then} you will not pass the course.
- d) $p \vee q \vee r$: You have the flu or you will miss the final examination or you will pass the course.
- e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$: Either if you have the flu, then you will not pass the course or if you miss the final examination, then you will not pass the course.
- f) $(p \wedge q) \vee (\neg q \wedge r)$: Either you have the flu and you ^{will} miss the final examination or you will not miss the final examination and you will pass the course.

15) Let p and q be the propositions p : You drive over 65 miles per hour.
 q : You get a speeding ticket.
 Write these propositions using p and q and logical connectives (including negations).

- a) You do not drive over 65 miles per hour — $\neg p$
- b) You drive over 65 miles per hour, but you do not get a speeding ticket — $p \wedge \neg q$.
- c) You will get a speeding ticket if you drive over 65 miles per hour — $p \rightarrow q$.
- d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket — $\neg p \rightarrow \neg q$.
- e) Driving over 65 miles per hour is sufficient for getting a speeding ticket. — $p \rightarrow q$.
- f) You get a speeding ticket, but you do not drive over 65 miles per hour — $q \wedge \neg p$.
- g) Whenever you get a speeding ticket, you are driving over 65 miles per hour — $q \rightarrow p$.